

DECISIVE MOMENT August 6, 1945 8:16 a.m.

Hiroshima

Around 7:15 am, Japanese radar detected three American aircraft flying south at high altitude. Because the planes were thought to be on reconnaissance, the air raid alert was lifted. At 8:16 am, the unchallenged leading B-29 bomber, named *Enola Gay*, dropped “Little Boy” over the city of Hiroshima. A nuclear weapon had been exploded in warfare for the first time.

The decision to bomb Hiroshima, and Nagasaki three days later, was made by US president Harry Truman. The war against Nazi Germany had ended in May, but Germany’s ally, Japan, refused to surrender. Any invasion of Japan would result in a huge loss of both military and civilian life, as well as massive physical destruction. Truman therefore decided to use atomic bombs to force Japan to surrender. Hiroshima was chosen because of its industrial and military significance. The Hiroshima bomb exploded 1,950 ft (600 meters) above the city with a blast equivalent to 13 kilotons of TNT. An estimated 90,000 people were killed instantly; another 50,000 died by the end of the year. About 90 percent of Hiroshima’s buildings were damaged or destroyed.

The physics that made this bomb possible grew out of the realization that the atom was not stable and indivisible but had the potential to release immense amounts of energy. In 1938, German scientists Otto Hahn and Fritz Strassmann split uranium atoms by bombarding them with neutrons. The process used, known as nuclear fission, had obvious

military uses, and scientists in the UK and US grew concerned that Germany might use it to make bombs. On August 2, 1939, Albert Einstein and the Hungarian physicist Leó Szilárd wrote to President Roosevelt urging him to take action. He set up the Uranium Committee to pursue research. After the US entered the war in December 1941, the “Manhattan Project,” under the direction of J. Robert Oppenheimer, developed a nuclear bomb.

Three devices resulted from this work: the “Trinity” test of a plutonium bomb detonated in Alamogordo, New Mexico; the enriched-uranium bomb detonated over Hiroshima; and the plutonium bomb dropped on Nagasaki. In 1949, the USSR exploded its first atomic bomb. Britain, France, and China soon followed. Today, Israel, India, Pakistan, and North Korea have joined the nuclear club. The threat of nuclear war might have acted as a deterrent and kept the peace between the main nuclear powers. The ability to destroy the world, however, has made our planet an infinitely more dangerous place.

Atomic cloud

The devastating force of the explosion threw a huge mushroom cloud thousands of feet into the sky. The searing heat of the bomb was so intense it vaporized people, and caused a firestorm that destroyed the city.

“My God, what have we done?”

ROBERT LEWIS, COPILOT OF THE *ENOLA GAY*, 1945